MB Note

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K})$ N: population at time t; K: carrying capacity (depends on resources); r,K: >0.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K+N_0(e^{rt}-1)} \overset{\infty}{\to} K$ **Equilibrium points:** $N=0$ (unstable), $N=K$ (stable). **Quickest:** $N = \frac{K}{2}$ (<: 凹; >: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T: delay time >0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

[1] **Ext. of Logistic Model:** $\frac{dN}{dt} = rN(t)\left[1-\frac{N(t-T)}{K}\right]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ [2] **Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*)\left[1-N^*(t^*-T^*)\right]$ ($N^*=\frac{N}{K}, t^*=rt, T^*=rT$)

[1,2] **Periodic sol:** stable limit cycle $N(t+t_p) = N(t), t_p$: period. **Amplitude:** rT **Equilibrium Points:** $N=0|N^*=0$ (unstable); $N=K|N^*=1$ (Stable: $0<T^*=rT<\frac{\pi}{2}$; Unstable: $T^*>\frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - EN = f(N)$ r,K,E: >0; EN: harvesting rate/unit time; E: effort measure.

Equilibrium: $N=0$ (unstable); $N=N_h(E) = K(1-\frac{E}{r})>0$ (need $E<r$, Stable) **Yield:** $Y(E)=EN$ $\cdot E>r$:die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E=\frac{r}{2}$) **Steady State:** $N_{h|Y_M} = \frac{K}{2}$

Continuous Population Model (Y_0): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0: constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** [1] $0<Y_0<Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: Verhulst: $N_{t+1} = rN_t(1-\frac{N_t}{K}); r, K>0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1-\frac{N_t}{K})}$

Imperfect Bifuraction: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r \le 0$, x^* unique. If $r>0$, $|h|>h_c$, x^* unique; $|h|<h_c$, x^* three. (两边 stable, 中 unstable). Bifuraction points: $h = \pm h_c = \pm\frac{2r}{3}\sqrt{\frac{r}{3}}$. **cusp point**: $(r,h)=(0,0)$

1.2 Summary - Stability and Bifuraction

	$\frac{dN}{dt} = f(N)$ **No Limit cycle periodic solutions** S: stable; U:unstable.				$N_{t+1} = N_t F(N_t) = f(N_t)$ **Eigenvalue:** $\lambda = f'(N^*)$			
Stability	Stable: $f'(N^*)<0$		Unstable: $f'(N^*)>0$		Stable: $\lvert\lambda\rvert<1$	Unstable: $\lvert\lambda\rvert>1$ (chaotic)	Periodic: $\lvert\lambda\rvert=1$ $\begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation}\end{cases}$	
Linearise	$N(t) = N^*+n(t), \lvert n(t)\rvert \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} =$ (general) $= f(N^*)+f'(N^*)n(t)$; (delay)= 直接带入 f 后近似				Let $N_t = N^*+n_t, \lvert n_t\rvert \ll 1 \Rightarrow N^*+n_{t+1} = \cdots = f(N^*+n_t) \approx f(N^*)+f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$			
Bifurcation \\ Perturbations at N^* | Periodic	Saddle-Note (Normal Form) $\dot{x} = r \pm x^2$ 0→S→U-S	Transcritical $\dot{x} = rx - x^2$ S-U→U-S	(super-)Pitchfork $\dot{x} = rx - x^3$ S→S-U-S	Sub-critical Pitchfork $\dot{x} = r + x^3$ U→U-S-U	$\lambda<-1$: ↑ oscillations	$-1<\lambda<0$: ↓ oscillations	$0<\lambda<1$: ↓ Monotone	$\lambda>1$: ↑ Monotone
					If $f^p(N^*) = N^*$, N^* is p-periodic. | $\forall$ even p-periodic N^* with $\lambda=-1$ branches into $2p$-periodic			
					Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \ge 1$, p-periodic N^* exists.			Species extinction: $N_{\min}<1$
Other	**Time-Independent**: At N^*: $\frac{dN}{dt}=0$ $N(t) = N(t-T) = N^*$ $f(N^*)=0$				**Max/Min**: [如果单峰 (unimodal)|2-periodic] $f'(N_m)=0$, $N_{\max} = f(N_m)$; $N_{\min} = f(N_{\max})$			
Other	**Tangency Condition**: If $\dot{x} = f(x,r)$, 分岔点 (x^*, r_c): $\frac{\partial f}{\partial x}\vert_{(x^*,r_c)} = 0$ $f(x^*,r_c)=0$ | 两点合并:coalesce.							

Hopf Bifuraction: 平衡点从稳定变为不稳定，同时产生 limit cycle. 条件: $\det(J)>0$; $\frac{d\text{tr}(J)}{d\mu} \neq 0$ or: tr(J) +/-符号有变化 $\Leftrightarrow$ $\lambda = \pm iw$ and $\frac{d\Re(\lambda)}{d\mu} \neq 0$ μ: parameter, not time.

$\Rightarrow$ Stable $\to$ Unstable + Stable Limit Cycle (supercritical) or Unstable $\to$ Stable + Unstable Limit Cycle (subcritical). (一般 super-) when tr$(J)=0|\Re(\lambda)=0$, $\lambda=\pm iw \Rightarrow$ **periodic:** $T=\frac{2\pi}{w}$

Bifuraction: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifuraction Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 |For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt}$ $\Rightarrow$ Assume $n(t) = e^{\lambda t}|ce^{\lambda t}$ $\to$ Find $\lambda = g(\lambda)$ $\Rightarrow$ Let $\lambda = u + iw \to u = g_1(u)$ and $w = g_2(w)$ $\Rightarrow$ $u<0$ asy.stable; $u>0$ unstable. **Find T_c:** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T=0$ stable. $\Rightarrow$ Using $u=0$ (i.e. $\lambda = iw$) to find T_c.

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a-bP)$; $\frac{dP}{dt} = P(cN-d)$ $\Rightarrow$ nondim: $\frac{du}{d\tau} = u(1-v)$; $\frac{dv}{d\tau} = \alpha v(u-1)$ **Equilibrium:** $(u^*,v^*) = (0,0),(1,1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H, H > H_{min} = \alpha+1$ constant. (closed phase) **Conservation Law:** Using $\frac{du}{dv} = \frac{du}{d\tau}\frac{d\tau}{dv}$ find H $\Rightarrow$ $\frac{dH}{d\tau} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u,v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv}\end{bmatrix}_{(u^*,v^*)}$ $\Rightarrow$ If $+/-$ definite (i.e. $\forall\lambda_i > 0 (<0)$)

$\Rightarrow$ It's local min/max at (u^*,v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u'(t) \\ v'(t)\end{bmatrix} = \begin{bmatrix} f(u,v) \\ g(u,v)\end{bmatrix}$ at equilibrium (u^*,v^*). Let $\begin{bmatrix} x \\ y\end{bmatrix} = \begin{bmatrix} u-u^* \\ v-v^*\end{bmatrix}$, then $\begin{bmatrix} x' \\ y'\end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v\end{bmatrix}_{(u^*,v^*)}\begin{bmatrix} x \\ y\end{bmatrix}$ $\Rightarrow$ Find eigenvalues λ of J by $\det(J-\lambda I)=0$

[1] $\det(J)>0, \text{tr}(J)<0 \Leftrightarrow \Re(\lambda)<0 \to$ stable node $_{Im(\lambda)=0}$; stable focus=spiral point $_{Im(\lambda)\neq 0}$; [2] $\det(J)>0, \text{tr}(J)>0 \Leftrightarrow \Re(\lambda)>0 \to$ unstable node $_{Im(\lambda)=0}$; unstable focus=spiral point $_{Im(\lambda)\neq 0}$

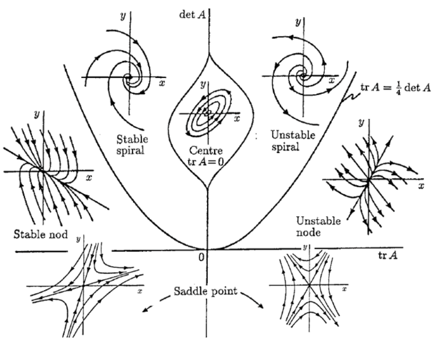
[3] $\det(J)<0 \Leftrightarrow Im(\lambda)=0, \lambda_1<0<\lambda_2 \to$ saddle point (unstable); [4] $\det(J)>0, \text{tr}(J)=0 \Leftrightarrow \lambda = \pm iw$ (pure imaginary) $\to$ center (only linear stability)|Hopf bifurcation (见前面)

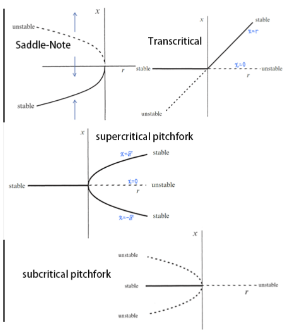
Nullclines: $f(u,v)=0$ or $g(u,v)=0$ $\to$ 得 u,v 表达线条是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有 边界的外法线方向 **n**, 在对应边界上的点, 满足 $\mathbf{n}\cdot\begin{bmatrix} f(u,v) \\ g(u,v)\end{bmatrix} < 0$

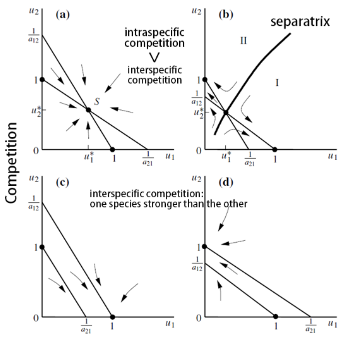
Find slope of Nullclines: $f(u,v)=0 \Rightarrow df = f_u du + f_v dv = 0 \Rightarrow \frac{dv}{du} = -\frac{f_u}{f_v}$ Similarly: $\frac{dv}{du} = -\frac{g_u}{g_v}$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be <0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d)







Mutualism|Symbiosis: (共生)

Case I: 图 (a) coexist: competition not aggressive.

Case II: 两条射线 both species grow unboundedly.(all unstable)

Case II: 两条向外的线相交在一点 both species tend to a stable coexistence.

For small perturbations: If fixed point is Center or Imporper Node (GM=1): Any small perturbation $\Rightarrow$ may change to spiral point. $\Rightarrow$ Not structurally stable.

Chaotic: Three dim or delay $\Rightarrow$ 可能 chaotic.

Limit Cycle: Any sufficiently perturbation from a stable limit cycle $\Rightarrow$ trajectory returns to the limit cycle. (i.e. tends to zero asymptotically with time)

3 Reaction Kinetics

Enzyme Kinetics: e.g. $S+E \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} SE \overset{k_2}{\to} P+E$ K_i: rate of reaction; S: substrate; E: enzyme; SE: complex; P: product. **Rate of uptake:** $v = \frac{d[P]}{dt}|_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g. $\alpha_1 A + \alpha_2 B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \beta_1 C + \beta_2 D \Rightarrow \text{rate}_{\text{forward}} = k_1[A]^{\alpha_1}[B]^{\alpha_2}$; $\text{rate}_{\text{backward}} = k_{-1}[C]^{\beta_1}[D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即 $\frac{d(\text{fast})}{dt} = 0$. (一般来说假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For $\frac{du}{d\tau} = f(u,v), \varepsilon\frac{dv}{d\tau} = g(u,v), 0<\varepsilon \ll 1$ $\Rightarrow$ v fast (when $g \not\to 0$), u slow. **Quasi-steady-state approximation:** $\varepsilon \to 0 \Rightarrow g(u,v)=0$

Taylor: $u(\tau,\varepsilon) = \sum_{n=0}^{\infty}\varepsilon^n u_n(\tau)$; $v(\tau,\varepsilon) = \sum_{n=0}^{\infty}\varepsilon^n v_n(\tau)$ **Outer solution:** Taylor 带入原方程组 $\to$ 保留 $\mathcal{O}(1)|\varepsilon=0$ Valid for $\tau>0$ (slow time); $v_0(0)$ **unvalid**.$\to$not uniformly valid for $\tau \ge 0$

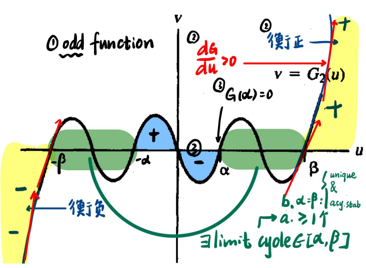
Inner solution: $\sigma = \frac{\tau}{\varepsilon}$, $\begin{matrix} U(\sigma;\varepsilon)=u(\tau;\varepsilon) \\ V(\sigma;\varepsilon)=v(\tau;\varepsilon)\end{matrix}$ $\Rightarrow$ $\frac{dU}{d\sigma} = \varepsilon f(u,v)$, $\frac{dV}{d\sigma} = g(u,v) \to$ Taylor 带入原方程组 $\to$ 保留 $\mathcal{O}(1)|\varepsilon=0$ (known as boundary layer ($\tau=0$ 处)); Valid $\tau=0$ (fast time)

Asymptotic Matching: $\lim_{\sigma\to\infty}[U_0(\sigma),V_0(\sigma)] = \lim_{\tau\to 0^+}[u_0(\tau),v_0(\tau)]$ **Uniformly Valid Solution:** $u(\tau;\varepsilon) = u_0(\tau)+\mathcal{O}(\varepsilon)$; $v(\tau;\varepsilon) = \begin{cases} v_0(\tau)+\mathcal{O}(\varepsilon), & 0<\varepsilon\ll\tau \\ V_0(\sigma)+\mathcal{O}(\varepsilon), & 0<\tau\ll 1\end{cases}$

Activator-Inhibitor Model: For $\frac{du}{dt} = f(u,v)$, $\frac{dv}{dt} = g(u,v)$: If $\frac{\partial g}{\partial u}>0$, u is an **activator** of v; If $\frac{\partial f}{\partial v}<0$, v is an **inhibitor** of u. ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: **Positive:** $f_u, f_v \ge 0; g_u, g_v \le 0$ or $f_u, f_v \le 0; g_u, g_v \ge 0$ **Negative:** $f_u, f_v \ge 0; g_u, g_v \ge 0$ or $f_u, f_v \le 0; g_u, g_v \le 0$

4 Relaxation Oscillations & Switch Behaviour

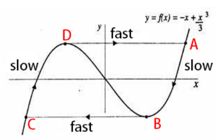


Find Limit Cycle: For $\begin{cases}\dot{u} = v - G(u) \\ \dot{v} = -u\end{cases}$ $\Leftrightarrow$ $\frac{d^2u}{dt^2} + g(u)\frac{du}{dt} + u = 0$ with $G(u) = \int_0^u g(s)ds$. Lienard Equations If we have:

1: $G(-u) = -G(u)$ (odd) 2: $\exists\alpha>0$ s.t. $G(\alpha)=0$ and $G(u)<0, \forall u \in (0,\alpha)$ 3: $\exists\beta \ge \alpha$ s.t. $G(u)>0$, $\frac{dG}{du}>0, \forall u>\beta$ and $G(u)\to\infty$ as $u\to\infty$.

Then: G is 'bumpy' function. [1] $\exists$ limit cycle between $(\alpha,\beta),(-\beta,-\alpha)$. [2] unique and asy.stable if $\alpha=\beta$

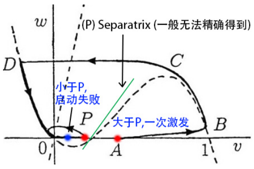
Relaxation Oscillations: A periodic movement of "slow-fast alternation". e.g. For $\begin{cases}\varepsilon\dot{u} = v - G(u) \\ \dot{v} = -u \mid -u\varepsilon\end{cases}$ or More general form: $\begin{cases}\varepsilon\dot{u} = f(u,v) \\ \dot{v} = g(u,v)\end{cases}$ $|\varepsilon| \ll 1$



Relaxation Oscillations requies unstable equilibrium point.

Explain why it occur: [1] Find *nullclines* (一般: 直线 +S 曲线) [2] Since $\dot{u} = \frac{\text{some func}}{\varepsilon} \gg 1$ (so that $|\dot{u}| \gg |\dot{v}|$) unless close to nullcline $\to$(一般 S 曲线) [3] 图上选 4 个点 ABCD (只是一个示例而已), 一般选 S 的 local max/min 加 S 更远处两个点, 连接成一个 loop. [4] 判断 ABCD 点的流向 (看 nullcline 两侧/上下) [5] 分段分析 + 从一不在 nullcline 上的点分析 e.g. $A \to B$ $\dot{u}=0, \dot{v}=-u<0$ 向下, 慢阶段. $B \to C$ not close to S curve, $|\dot{u}| \gg 1$ 向左, 快阶段/fast horizontal jump. $C \to D$ ... $D \to A$ ...

Estimate period T: 例如上图 (例子忽略了快反应) $T=\int_{t_A}^{t_B} dt+\int_{t_C}^{t_D} dt = 2\int_{t_A}^{t_B} dt$ **Find $dt = \_du$ 带入计算** e.g. In slow time: $\dot{u}=0 \to v=G(u)$ and $\dot{v}=-u \to \frac{dv}{dt}=\frac{dv}{du}\frac{du}{dt}=G'(u)\frac{du}{dt}=-u \to -\frac{G'(u)}{u}du=dt$, 带入计算



Excitable System: FitzHugh-Nagumo Model

$$\begin{cases} \frac{dx}{dt}=c[y+x-\frac{x^3}{3}+z(t)] \\ \frac{dy}{dt}=\frac{-x+a+by}{c} \end{cases}$$

(x: excitability variable; y: recovery variable; $z(t)$: external stimulus)

Threshold 现象: **Small perturbation**: return to equilibrium (failed initiation/excitation). **Large perturbation**: large excursion before returning to equili. (action potential|successful initiation). After excitation| 大扰动之后: Refractory Period不应期: system is unresponsive to further stimulation. **constant stimulus**: periodic firing (limit cycle).

Reason: 寻找 $\dot{u}$, $\dot{v}$ s.t. $\ll, \gg 1$ i.e. $|\dot{u}| \gg |\dot{v}|$ or $|\dot{u}| \ll |\dot{v}|$

Switch Behaviour: 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如有两 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

描述话语 example: 快-慢震荡: [1](option) The equilibrium point S is unstable between the two *knees/turnning points* of the u-nullcline/cubic-like curve. | or: S changes stability as it crosses the knees of the u-nullcline. [2] 图上标/简单解释拐点位置. [3] 证明 $|u'| \gg 1$(写成乘积形式判断)或其他 $\to$ 得快运动边. [4] Trajectories *starting off* the u-nullcline *repidly/quickly* approach *left/right branch* of the u-nullcline, and then *slowly move down/up|crawl along* the u-nullcline due to $v' < 0/ > 0$, and to *upper/lower knee/turnning point* [5] At the knee/turnning point, the trajectory *quickly jumps* to the *right/left branch* of the u-nullcline, and then *slowly moves up/down|crawl along* the u-nullcline due to $v' > 0/ < 0$. [6] And this process *repeats periodically*, giving rise to a *relaxation oscillation*. **Sxcitable**: 类似的, 分析 small perturbation *above/below* equilibrium point S -> 得 *failed excitation* or *successful excitation*.

5 Appendix

$y'+p(x)y=q(x)$: **Integrating Factor**: $I(x)=e^{\int p(x)dx} \Rightarrow I(x)y=\int I(x)q(x)dx$ **Taylor Expansion:** $f(x,r)$ at (x_0,r_0): $f(x,r)=f(x_0,r_0)+(x-x_0)f_x|_{(x_0,r_0)}+(r-r_0)f_r|_{(x_0,r_0)}+\frac{1}{2}\cdots$

Binomial Series: $(1+x)^\alpha = 1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^2+\cdots$ **Linerisation:** $f(x+\Delta x)=f(x)+f'(x)\Delta x+\cdots$ $e^{bx}=1+bx+\frac{(bx)^2}{2!}+\cdots$ $\ln(a+x)=\ln(a)+\frac{x}{a}-\frac{x^2}{2a^2}+\frac{x^3}{3a^3}-\cdots$

Hyperbolic Functions: $\sinh x=\frac{e^x-e^{-x}}{2}$ $\cosh x=\frac{e^x+e^{-x}}{2}$ $\tanh x=\frac{\sinh x}{\cosh x}$ $\sinh(x)=x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots$ $\cosh(x)=1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$ $\tanh(x)=x-\frac{x^3}{3}+\frac{2x^5}{15}-\cdots$